

PRATIKUM MODUL 1

PENGEMBANGAN UI LANJUT

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Struktur Aplikasi dan Navigasi Dalam Vue.js

Week 2

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1. TUJUAN

Memahami serta menerapkan keterkaitan Site Map, Views, dan Router, membangun hierarki halaman Parent Child merancang transisi navigasi interaktif Navbar dengan routerlink serta membuat penunjuk posisi (Breadcrumb) pada aplikasi web.

2. ALAT DAN BAHAN

- a. Komputer atau Laptop
- b. Visual Studio Code
- c. Vue.js
- d. Node.js

3. KODE DAN PENJELASAN

A. Inisialisasi dan Kerangka Dasar

a. src/main.js

Merupakan entry point utama aplikasi Vue. File ini bertugas membuat instance aplikasi, mendaftarkan modul state management dan sistem navigasi.

```
belajar-vue > src > JS main.js > ...
1  import './assets/main.css'
2
3  import { createApp } from 'vue'
4  import { createPinia } from 'pinia'
5
6  import App from './App.vue'
7  import router from './router'
8
9  const app = createApp(App)
10
11 app.use(createPinia())
12 app.use(router)
13
14 app.mount('#app')
```

b. src/App.vue

Berfungsi sebagai root component dasar. Berisi elemen yang bertindak sebagai placeholder untuk menampilkan komponen halaman sesuai dengan rute URL yang sedang aktif.

```
belajar-vue > src > App.vue > ...
1  <script setup>
2  import { RouterView } from 'vue-router'
3  </script>
4
5  <template>
6  |   <RouterView />
7  </template>
```

c. src/layouts/App.vue

Sebagai layout global aplikasi. Menggabungkan komponen navigasi atas (Navbar), penunjuk posisi (Breadcrumb), dan di dalam kontainer utama berformat responsif agar struktur navigasi tetap konsisten di setiap halaman.

```
belajar-vue > src > layouts > ▼ App.vue > ...
1  <script setup lang="ts">
2  import Navbar from '@components/app/Navbar.vue';
3  import Breadcrumb from '@components/app/Breadcrumb.vue';
4  </script>
5
6  <template>
7    <div class="app-layout">
8      <Navbar />
9      <main class="main-content">
10       <Breadcrumb />
11       <router-view />
12     </main>
13   </div>
14 </template>
15
16 <style scoped>
17 .app-layout {
18   min-height: 100vh;
19   display: flex;
20   flex-direction: column;
21 }
22 .main-content {
23   flex: 1;
24   padding: 2rem;
25   max-width: 1440px;
26   margin: 0 auto;
27   width: 100%;
28 }
29 </style>
```

B. Mengatur Rute

a. src/router/index.js

Menginisialisasi Vue Router menggunakan `createWebHistory` dan mendefinisikan pemetaan URL ke komponen tertentu (Home, About, Contact, Browse).

```
belajar-vue > src > router > JS index.js > ...
4  const routes = [
5    {
6      path: '/',
7      component: AppLayout,
8      children: [
9        {
10         path: '',
11         name: 'home',
12         component: () => import('@/views/Home.vue'),
13         meta: { breadcrumb: 'Home' },
14       },
15       {
16         path: 'about',
17         name: 'about',
18         component: () => import('@/views/About.vue'),
19         meta: { breadcrumb: 'About' },
20       },
21       {
22         path: 'browse',
23         name: 'browse',
24         component: () => import('@/views/Browse.vue'),
25         meta: { breadcrumb: 'Browse' },
26         redirect: '/browse/events',
27         children: [
28           {
29             path: 'events',
30             name: 'events',
31             component: () => import('@/views/EventList.vue'),
32             meta: { breadcrumb: 'Event List' },
33           },
34           {
35             path: 'events/:id',
36             name: 'event-detail',
37             component: () => import('@/views/EventDetail.vue'),
38             meta: { breadcrumb: 'Event Detail' },
39           },
40           {
41             path: 'category',
42             name: 'category',
43             component: () => import('@/views/Category.vue'),
44             meta: { breadcrumb: 'Category' },
45           },
46         ],
47       },
48     ],
49     {
50       path: 'contact',
51       name: 'contact',
52       component: () => import('@/views/Contact.vue'),
53       meta: { breadcrumb: 'Contact' },
54     },
55   ],
56 ]
57
58 const router = createRouter({
59   history: createWebHistory(import.meta.env.BASE_URL),
60   routes,
61 })
62
63 export default router
```

C. Transisi dan Orientasi (Navbar & Breadcrumb)

a. src/components/app/Navbar.vue

Mengelola navigasi utama sebagai instrumen transisi antarhalaman.

```
1  <script setup>
2  import { ref, onMounted, onUnmounted } from 'vue'
3  import { useRoute } from 'vue-router'
4
5  const isScrolled = ref(false)
6  const route = useRoute()
7
8  const menus = [
9    { name: 'Home', path: '/' },
10   { name: 'About', path: '/about' },
11   {
12     name: 'Browse',
13     path: '/browse',
14     children: [
15       {
16         name: 'Event List',
17         path: '/browse/events',
18         children: [
19           {
20             name: 'Event Detail (Sample)',
21             path: '/browse/events/1'
22           }
23         ]
24       },
25       {
26         name: 'Category',
27         path: '/browse/category'
28       }
29     ]
30   },
31   { name: 'Contact', path: '/contact' }
32 ]
33
34 const handleScroll = () => {
35   isScrolled.value = window.scrollY > 10
36 }
37
38 onMounted(() => {
39   window.addEventListener('scroll', handleScroll)
40 })
41
42 onUnmounted(() => {
43   window.removeEventListener('scroll', handleScroll)
44 })
45 </script>
46
47 <template>
48   <nav class="navbar" :class="{ scrolled: isScrolled }">
49     <div class="navbar-container">
50
51       <!-- Logo -->
52       <router-link to="/" class="logo">
53         <svg
54           width="28"
55           height="32"
56           viewBox="0 0 24 28"
57           fill="none"
58           xmlns="http://www.w3.org/2000/svg"
59           class="logo-icon"
60         >
```

```

61     <path
62       d="M12 0L22.3923 6V18L12 24L1.6077 18V6L12 0Z"
63       fill="white"
64     />
65
66     <path
67       d="M15 9.5 C15 9.5 14 8 12 8 C9.5 8 8 10 8 12.5 C8 15 9.5 17 12 17 C14 17 15 16 15.5 14.5 V 12.5 H 12.5"
68       stroke="#1A1643"
69       stroke-width="2.5"
70       stroke-linecap="round"
71       stroke-linejoin="round"
72     />
73   </svg>
74
75   <span class="logo-text">Gatherly</span>
76 </router-link>
77
78 <!-- Navigation Menu -->
79 <ul class="nav-menu">
80   <li
81     v-for="menu in menus"
82     :key="menu.name"
83     class="nav-item"
84   >
85     <router-link
86       :to="menu.path"
87       class="nav-link"
88       :class="{
89         active:
90           route.path === menu.path ||
91           (menu.path !== '/' && route.path.startsWith(menu.path))
92       }"
93     >
94       {{ menu.name }}
95
96       <svg
97         v-if="menu.children"
98         class="dropdown-indicator"
99         width="12"
100        height="12"
101        viewBox="0 0 24 24"
102        fill="none"
103        stroke="currentColor"
104        stroke-width="2"
105        stroke-linecap="round"
106        stroke-linejoin="round"
107      >
108        <polyline points="6 9 12 15 18 9" />
109      </svg>
110    </router-link>

```

```

110 </router-link>
111
112 <!-- First Level Dropdown -->
113 <ul v-if="menu.children" class="dropdown-menu">
114   <li
115     v-for="child in menu.children"
116     :key="child.name"
117     class="dropdown-item"
118   >
119     <router-link
120       :to="child.path"
121       class="dropdown-link"
122     >
123       {{ child.name }}
124
125       <svg
126         v-if="child.children"
127         class="submenu-indicator"
128         width="12"
129         height="12"
130         viewBox="0 0 24 24"
131         fill="none"
132         stroke="currentColor"
133         stroke-width="2"
134         stroke-linecap="round"
135         stroke-linejoin="round"
136       >
137         <polyline points="9 18 15 12 9 6" />
138       </svg>
139     </router-link>
140
141     <!-- Second Level Dropdown -->
142     <ul v-if="child.children" class="submenu">
143       <li
144         v-for="subchild in child.children"
145         :key="subchild.name"
146         class="submenu-item"
147       >
148         <router-link
149           :to="subchild.path"
150           class="dropdown-link"
151         >
152           {{ subchild.name }}
153         </router-link>
154       </li>
155     </ul>
156   </li>
157 </ul>
158 </li>
159 </ul>
160

```

```

161 <!-- Right Navigation -->
162 <div class="nav-right">
163
164 <!-- Language Selector -->
165 <div class="lang-selector">
166 <svg
167   width="20"
168   height="20"
169   viewBox="0 0 24 24"
170   fill="none"
171   stroke="currentColor"
172   stroke-width="2"
173   stroke-linecap="round"
174   stroke-linejoin="round"
175 >
176   <circle
177     cx="12"
178     cy="12"
179     r="10"
180   />
181
182   <line
183     x1="2"
184     y1="12"
185     x2="22"
186     y2="12"
187   />
188
189   <path
190     d="M12 2a15.3 15.3 0 0 1 4 10 15.3 15.3 0 0 1 4 10 15.3 15.3 0 0 1 4 10 15.3 15.3 0 0 1 4 10z"
191   />
192 </svg>
193
194 <span class="lang-text">EN</span>
195
196 <svg
197   width="18"
198   height="18"
199   viewBox="0 0 24 24"
200   fill="none"
201   stroke="currentColor"
202   stroke-width="2"
203   stroke-linecap="round"
204   stroke-linejoin="round"
205   class="chevron"
206 >
207   <polyline points="6 9 12 15 18 9" />
208 </svg>
209 </div>
210

```

```

211 <!-- Hamburger -->
212 <button class="hamburger-btn">
213   <svg
214     width="24"
215     height="24"
216     viewBox="0 0 24 24"
217     fill="none"
218     stroke="currentColor"
219     stroke-width="1.5"
220     stroke-linecap="round"
221     stroke-linejoin="round"
222   >
223     <line x1="4" y1="12" x2="20" y2="12"></line>
224     <line x1="4" y1="6" x2="20" y2="6"></line>
225     <line x1="4" y1="18" x2="20" y2="18"></line>
226   </svg>
227 </button>
228
229 </div>
230 </div>
231 </nav>
232 </template>

```

b. src/components/app/Breadcrumb.vue

Berfungsi sebagai instrumen orientasi pengguna (location indicator). Mengubah parameter dinamis pada rute detail menjadi URL penuh serta memastikan rute dasar (Home) selalu berada di posisi terdepan. Menampilkan tautan aktif dan pemisah (/) secara visual di antarmuka.

```
belajar-vue > src > components > app > ▼ Breadcrumb.vue > ...
1  <script setup>
2  import { computed } from 'vue';
3  import { useRoute } from 'vue-router';
4
5  const route = useRoute();
6
7  const breadcrumbs = computed(() => {
8    const matched = route.matched;
9    let crumbs = matched.map(m => {
10      let path = m.path;
11      if (path.includes(':')) {
12        path = route.path;
13      }
14      return {
15        name: m.name,
16        path: path,
17        meta: m.meta
18      };
19    }).filter(m => m.meta && m.meta.breadcrumb);
20
21    if (route.name === 'event-detail') {
22      crumbs.splice(crumbs.length - 1, 0, {
23        path: '/browse/events',
24        meta: { breadcrumb: 'Event List' }
25      });
26    }
27
28    // Prepend Beranda if it's not already the first item
29    if (crumbs.length === 0 || crumbs[0].meta.breadcrumb !== 'Home') {
30      crumbs.unshift({
31        path: '/',
32        meta: { breadcrumb: 'Home' }
33      });
34    }
35
36    return crumbs;
37  });
38 </script>
39
40 <template>
41   <nav class="breadcrumb" v-if="breadcrumbs.length > 0">
42     <ul>
43       <li v-for="(crumb, index) in breadcrumbs" :key="index">
44         <span v-if="index > 0" class="separator"></span>
45         <router-link v-if="index < breadcrumbs.length - 1" :to="crumb.path">
46           {{ crumb.meta.breadcrumb }}
47         </router-link>
48         <span v-else class="active-crumb">{{ crumb.meta.breadcrumb }}</span>
49       </li>
50     </ul>
51   </nav>
52 </template>
```

D. Destinasi Halaman Pendukung

a. src/views/Home.vue

Merupakan halaman awal yang memuat hero section, ringkasan fitur utama, serta visualisasi struktur situs untuk menggambarkan arsitektur rute Vue Router.

```
belajar-vue > src > views > Home.vue > ...
1  <template>
2  <div class="home-page">
3    <section class="hero">
4      <div class="hero-content">
5        <span class="badge">Welcome to Gatherly</span>
6        <h1 class="hero-title">Connect, Discover, and Experience</h1>
7        <p class="hero-subtitle">
8          Join our vibrant community to explore the best events tailored for you.
9          Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor
10         incididunt ut labore et dolore magna aliqua.
11       </p>
12       <div class="hero-actions">
13         <router-link to="/browse/events" class="btn btn-primary">Discover Events</router-link>
14         <router-link to="/about" class="btn btn-secondary">Learn More</router-link>
15       </div>
16     </div>
17   </section>
18
19   <section class="features">
20     <div class="feature-card" v-for="i in 3" :key="i">
21       <div class="feature-icon">{{ ['?', '??', '???'][i-1] }}</div>
22       <h3>{{ ['Curated Events', 'Global Reach', 'Community Driven'][i-1] }}</h3>
23       <p>Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea
24         commodo consequat.</p>
25     </div>
26   </section>
27
28   <section class="sitemap-visual">
29     <h2>Website Structure / Site-Map</h2>
30     <p class="sitemap-desc">A quick overview of how this application is structured via Vue Router.</p>
31     <ul class="tree">
32       <li>
33         <router-link to="/">Home</router-link>
34       </li>
35       <li><router-link to="/about">About</router-link></li>
36       <li>
37         <router-link to="/browse">Browse</router-link>
38         <ul>
39           <li>
40             <router-link to="/browse/events">Event List</router-link>
41             <ul>
42               <li><router-link to="/browse/events/1">Event Detail (Example)</router-link></li>
43             </ul>
44           </li>
45           <li><router-link to="/browse/category">Category</router-link></li>
46         </ul>
47       </li>
48       <li><router-link to="/contact">Contact</router-link></li>
49     </ul>
50   </section>
51 </div>
52 </template>
```

b. src/views/About.vue

Merepresentasikan perpindahan antarmenu utama pada tingkat hierarki yang sama. About.vue berisi profil, visi, dan misi platform.

```
belajar-vue > src > views > About.vue > ...
1  <template>
2    <div class="about-page">
3      <div class="about-header">
4        <h1>About Gatherly</h1>
5        <p>Building the future of event experiences, one community at a time.</p>
6      </div>
7
8      <div class="about-content">
9        <div class="text-block">
10         <h2>Our Mission</h2>
11         <p>
12           Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat
13           nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia
14           deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis iste natus error sit
15           voluptatem accusantium doloremque laudantium.
16         </p>
17       </div>
18
19       <div class="text-block">
20         <h2>Our Vision</h2>
21         <p>
22           Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer nec odio. Praesent
23           libero. Sed cursus ante dapibus diam. Sed nisi. Nulla quis sem at nibh elementum
24           imperdiet. Duis sagittis ipsum. Praesent mauris. Fusce nec tellus sed augue semper porta.
25         </p>
26       </div>
27     </div>
28   </div>
29 </template>
```

c. src/views/Contact.vue

Merepresentasikan perpindahan antarmenu utama pada tingkat hierarki yang sama. sedangkan Contact.vue menyediakan informasi kontak dan formulir pengiriman pesan.

```
belajar-vue > src > views > Contact.vue > ...
1  <template>
2  <div class="contact-page">
3    <div class="contact-header">
4      <h1>Contact Us</h1>
5      <p>
6        We'd love to hear from you. Please fill out the form below or reach out via our contact
7        info.
8      </p>
9    </div>
10
11    <div class="contact-content">
12      <div class="contact-info">
13        <div class="info-item">
14          <div class="info-icon">??</div>
15          <div>
16            <h3>Our Office</h3>
17            <p>123 Event Street, Tech City, 10101</p>
18          </div>
19        </div>
20        <div class="info-item">
21          <div class="info-icon">??</div>
22          <div>
23            <h3>Phone</h3>
24            <p>+62 812 3456 7890</p>
25          </div>
26        </div>
27        <div class="info-item">
28          <div class="info-icon">??</div>
29          <div>
30            <h3>Email</h3>
31            <p>hello@gatherly.com</p>
32          </div>
33        </div>
34      </div>
35
36      <div class="contact-form">
37        <form @submit.prevent>
38          <div class="form-group">
39            <label>Name</label>
40            <input type="text" placeholder="John Doe" class="form-control" />
41          </div>
42          <div class="form-group">
43            <label>Email</label>
44            <input type="email" placeholder="john@example.com" class="form-control" />
45          </div>
46          <div class="form-group">
47            <label>Message</label>
48            <textarea placeholder="How can we help you?" rows="5" class="form-control"></textarea>
49          </div>
50          <button class="btn-submit">Send Message</button>
51        </form>
52      </div>
53    </div>
54  </div>
55 </template>
```

d. src/views/Browse.vue

Merupakan komponen wrapper untuk route untuk menampilkan halaman turunan child routes tanpa mengubah kerangka utama.

```
belajar-vue > src > views > Browse.vue > ...
1  <template>
2    <div class="browse-page">
3      <div class="browse-header">
4        <h1>Browse & Discover</h1>
5        <p>Find the events and categories that matter most to you.</p>
6      </div>
7      <div class="browse-content">
8        <router-view></router-view>
9      </div>
10   </div>
11 </template>
```

e. src/views/Category.vue

Merupakan halaman turunan dari Browse yang menampilkan klasifikasi acara berdasarkan kategori menggunakan tata letak berbasis grid.

```
belajar-vue > src > views > Category.vue > ...
1  <template>
2    <div class="category-page">
3      <h2 class="section-title">Explore by Category</h2>
4      <div class="category-grid">
5        <div class="category-card" v-for="(cat, index) in categories" :key="index">
6          <div class="cat-icon">{{ cat.icon }}</div>
7          <h3>{{ cat.name }}</h3>
8          <p>{{ cat.count }} Events</p>
9        </div>
10     </div>
11   </div>
12 </template>
13
14 <script setup>
15   const categories = [
16     { name: 'Music & Concerts', icon: '??', count: 24 },
17     { name: 'Technology', icon: '??', count: 18 },
18     { name: 'Art & Design', icon: '??', count: 12 },
19     { name: 'Business', icon: '??', count: 30 },
20     { name: 'Health & Wellness', icon: '??', count: 15 },
21     { name: 'Food & Drink', icon: '??', count: 22 },
22   ]
23 </script>
```

f. src/views/EventList.vue

Menampilkan daftar katalog acara dan menyediakan tautan rute dinamis menuju halaman rincian masing-masing acara.

```
belajar-vue > src > views > EventList.vue > ...
1 <template>
2   <div class="event-list-page">
3     <div class="header-section">
4       <h2 class="section-title">Upcoming Events</h2>
5       <p class="section-desc">Discover the latest gatherings and activities happening near you.</p>
6     </div>
7
8     <div class="event-grid">
9       <div class="event-card" v-for="i in 6" :key="i">
10         <div class="event-body">
11           <div class="event-meta">
12             <span class="event-date">Oct {{ 10 + i }}, 2026</span>
13             <span class="event-category">Community</span>
14           </div>
15           <h3>Community Gathering {{ i }}</h3>
16           <p class="event-loc">?? Main Auditorium, City Center</p>
17           <p class="event-desc">
18             Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aliquam at velit vel magna
19             interdum scelerisque.
20           </p>
21           <div class="card-footer">
22             <router-link :to="'/browse/events/${i}'" class="btn-link">
23               View Details &arr;
24             </router-link>
25           </div>
26         </div>
27       </div>
28     </div>
29   </div>
30 </template>
```

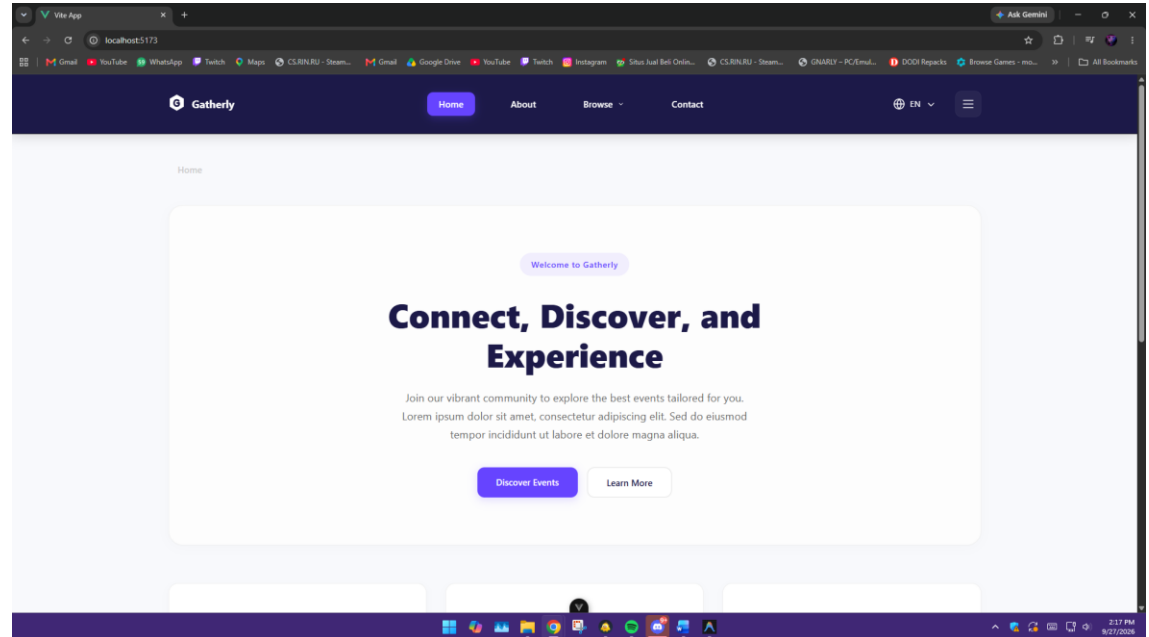
g. src/views/EventDetail.vue

Merupakan tingkat terdalam navigasi yang menampilkan data berdasarkan parameter URL dinamis. Halaman ini juga menyediakan navigasi kembali menggunakan serta menampilkan rincian agenda dan registrasi tiket.

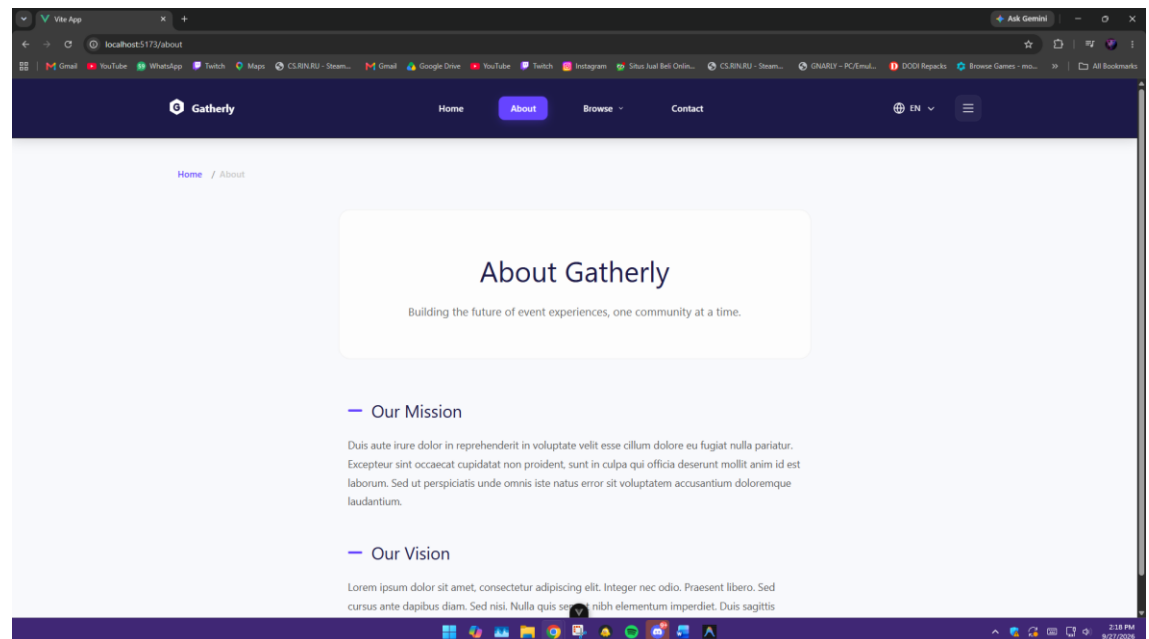
```
belajar-vue > src > views > EventDetail.vue > ...
1  <template>
2    <div class="event-detail-page">
3      <button @click="$router.push('/browse/events')" class="btn-back">&larr; Back to Events</button>
4
5      <div class="detail-header">
6        <div class="header-content">
7          <span class="event-tag">Community Event</span>
8          <h1>Community Gathering {{ $route.params.id }}</h1>
9          <div class="meta-info">
10             <div class="meta-item">
11               <span class="icon">??</span>
12               <span>October 15, 2026</span>
13             </div>
14             <div class="meta-item">
15               <span class="icon">??</span>
16               <span>Main Auditorium, City Center</span>
17             </div>
18             <div class="meta-item">
19               <span class="icon">??</span>
20               <span>500+ Attendees</span>
21             </div>
22           </div>
23         </div>
24       </div>
25
26       <div class="detail-content">
27         <div class="main-desc">
28           <h2>About This Event</h2>
29           <p>
30             Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nullam in dui mauris. Vivamus hendrerit arcu
31             sed erat molestie vehicula. Sed auctor neque eu tellus rhoncus ut eleifend nibh porttitor. Ut in
32             nulla enim. Phasellus molestie magna non est bibendum non venenatis nisl tempor.
33           </p>
34           <p>
35             Suspendisse dictum feugiat nisl ut dapibus. Mauris iaculis porttitor posuere. Praesent id
36             metus massa, ut blandit odio. Proin quis tortor orci. Etiam at risus et justo dignissim
37             congue. Donec congue lacinia dui, a porttitor lectus condimentum laoreet. Nunc eu ullamcorper orci.
38           </p>
39         </div>
40         <h2>Agenda</h2>
41         <ul class="agenda-list">
42           <li><strong>09:00 AM</strong> - Registration & Welcome Coffee</li>
43           <li><strong>10:00 AM</strong> - Opening Keynote Speech</li>
44           <li><strong>11:30 AM</strong> - Networking Session</li>
45           <li><strong>01:00 PM</strong> - Closing Remarks</li>
46         </ul>
47       </div>
48       <div class="sidebar">
49         <div class="ticket-card">
50           <h3>Registration</h3>
51           <p class="price">Free</p>
52           <p class="ticket-desc">Secure your spot today before it runs out.</p>
53           <button class="btn-register">Register Now</button>
54           <p class="spots">Only 45 spots left!</p>
55         </div>
56       </div>
57     </div>
58   </template>
59
```

4. SCREENSHOT HASIL PRAKTIK

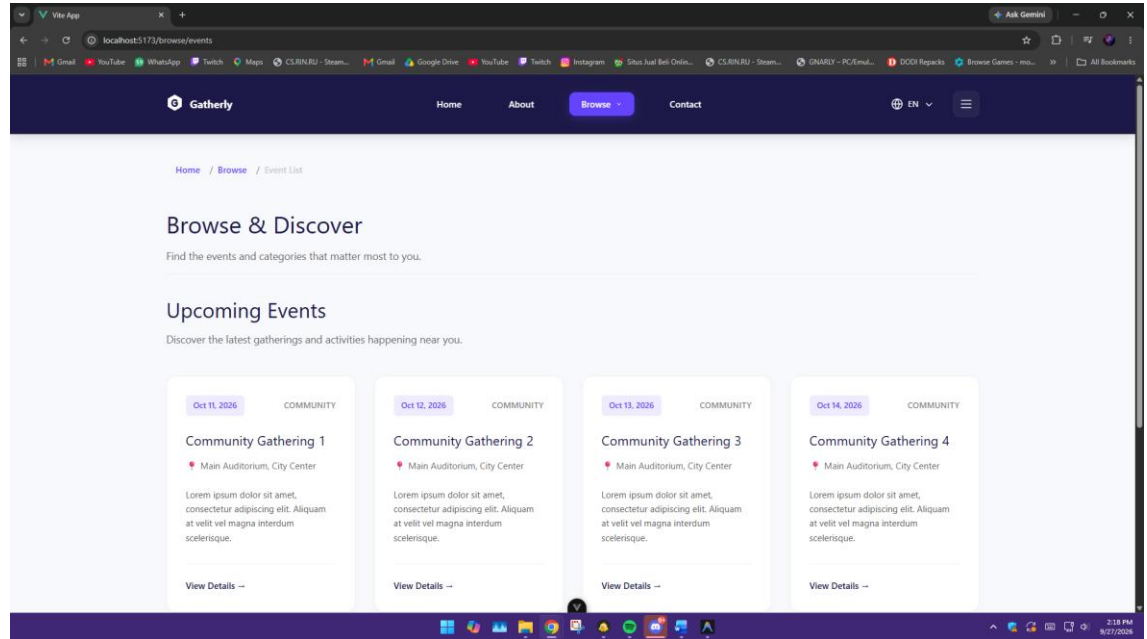
a. Tampilan home



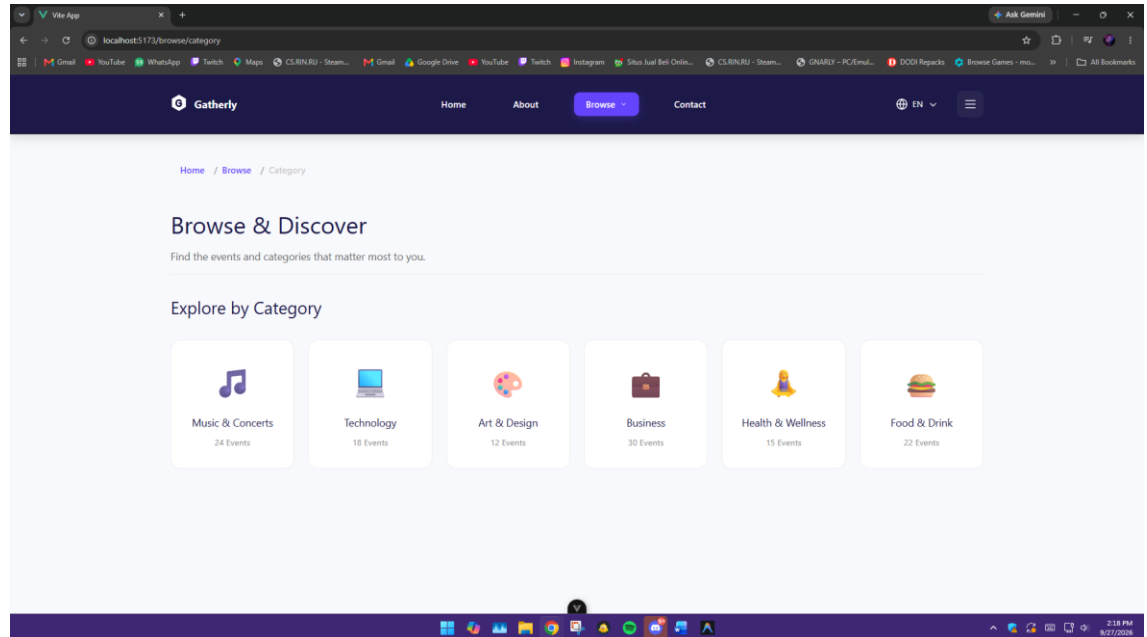
b. /about



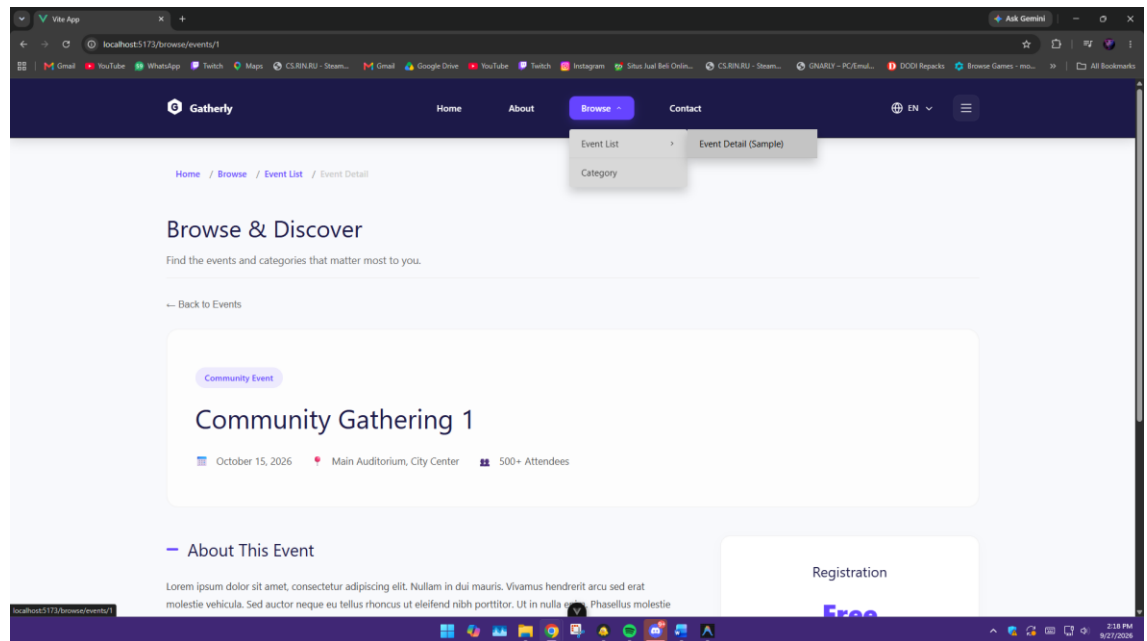
c. `/browse/events`



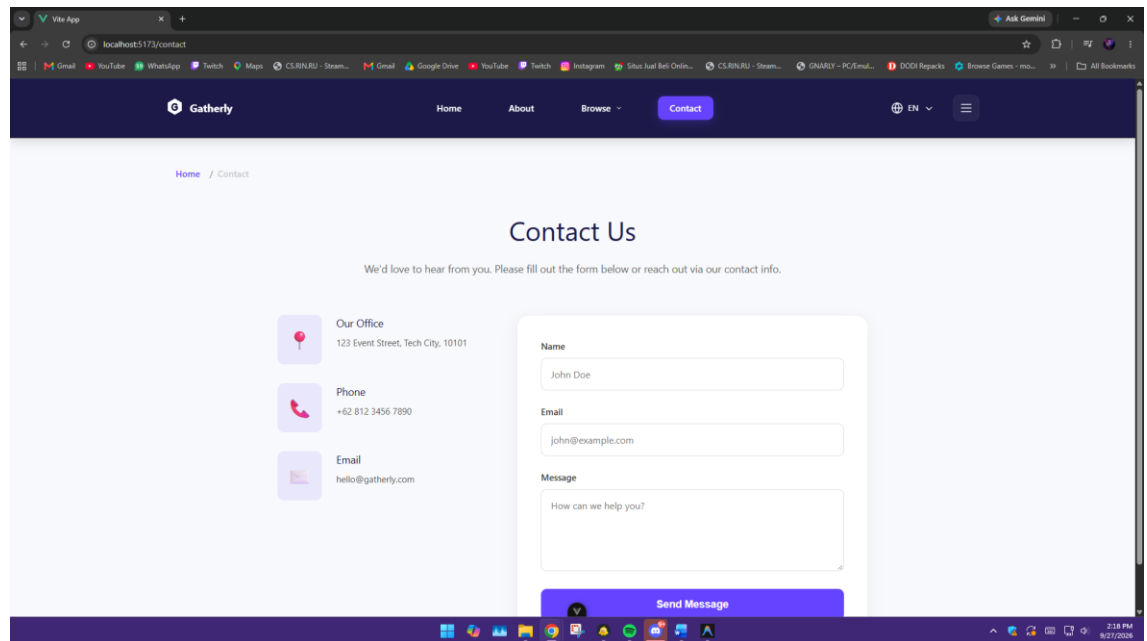
d. `/browse/category`



e. /browse/events/1



f. /contact



5. KESIMPULAN

Berdasarkan praktikum yang telah dilaksanakan, dapat disimpulkan bahwa penerapan arsitektur aplikasi menggunakan Vue.js berhasil mengoptimalkan pengelolaan struktur website serta alur navigasi antarmuka tanpa memerlukan pemuatan ulang halaman secara penuh. Integrasi hierarki rute parent child dan dynamic route yang dipadukan dengan komponen Navbar sebagai instrumen transisi serta Breadcrumb sebagai instrumen orientasi terbukti mampu menciptakan alur pergerakan pengguna secara terstruktur, efisien, dan responsif sesuai standar perancangan antarmuka modern.

6. LINK GITHUB

<https://github.com/enricostevenernest/belajar-vue>